

Group 1: Pipelining (Instruction-Level Parallelism)

- **What their slides must cover:** The basic fetch-decode-execute pipeline stages, latch registers, and pipeline execution timing charts (space-time diagrams).
- **Instructor Q&A Questions:**
- *"What happens to the entire pipeline when a conditional branch instruction occurs, and how do modern processors mitigate this?"* (Looking for: Pipeline stalls/flushes, branch prediction).
- *"If a 5-stage pipeline has a clock cycle of 2ns, why doesn't it actually achieve a perfect 5x speedup in practice?"* (Looking for: Pipeline overhead, structural/data hazards).

Group 2: Superscalar Architectures (Instruction-Level Parallelism)

- **What their slides must cover:** Multiple execution units (ALUs), the role of the instruction issue unit, out-of-order execution, and the difference between superscalar and deep pipelining.
- **Instructor Q&A Questions:**
- *"What is the difference between an in-order issue with out-of-order execution versus an out-of-order issue?"* (Looking for: Instruction scheduling complexity, RAW/WAR/WAW dependencies).
- *"Why is the instruction decode and issue unit often the biggest bottleneck in a superscalar processor?"* (Looking for: The complexity of analyzing multiple instructions for dependencies simultaneously in